

TransitOps

Smart Transport Operations Platform

TransitOps Team

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- 1. Executive Summary
- 2. Problem Statement
- 3. Product Vision & Differentiation
 - 3.1 Vision
 - 3.2 How this differs from Ecount-style tools
 - 3.3 USP Statement
- 4. Target Users & Personas
- 5. Scope
 - 5.1 In Scope (MVP — must ship in 8 hours)
 - 5.2 In Scope (Enhancement — build if time allows, after MVP is stable)
 - 5.3 Out of Scope / Stretch (post-hackathon roadmap)
- 6. Functional Requirements
 - 6.1 Authentication & RBAC
 - 6.2 Dashboard
 - 6.3 Vehicle Registry
 - 6.4 Driver Management
 - 6.5 Trip Management
 - 6.6 Maintenance
 - 6.7 Fuel & Expense Management
 - 6.8 Reports & Analytics
 - 6.9 Settings & RBAC
- 7. Business Rules (Mandatory + Derived)
- 8. Non-Functional Requirements
- 9. Data Entities (Overview)
- 10. Acceptance Criteria (Sample — Core Flow)
- 11. MVP Prioritization (MoSCoW — for the 8-hour build)
- 12. Risks & Mitigations

1. Executive Summary

TransitOps is a centralized, role-based transport operations platform that digitizes the full lifecycle of fleet management for logistics companies — from vehicle registration and driver compliance to trip dispatch, maintenance, fuel/expense tracking, and financial analytics.

Most SMB logistics operators today run this entire workflow on spreadsheets and paper logbooks (or on generic tools like Ecount), leading to double-booked vehicles, missed maintenance, expired driver licenses going unnoticed, and no real visibility into per-vehicle profitability. TransitOps replaces that with one login-gated system where every action — dispatching a trip, logging fuel, closing a maintenance ticket — automatically updates vehicle/driver status and rolls up into cost and performance analytics.

This document defines the product scope, functional requirements, business rules, and MVP prioritization for an 8-hour build.

2. Problem Statement

Transport and logistics SMBs typically manage fleet operations through disconnected tools: WhatsApp for dispatch, Excel for fuel logs, paper registers for maintenance, and manual reminders (or none) for license/document expiry. This causes:

- Vehicles or drivers double-booked onto overlapping trips
- Suspended or license-expired drivers still being assigned
- Overloaded vehicles (cargo exceeding rated capacity)
- Missed maintenance leading to breakdowns
- No visibility into which vehicle is actually profitable (fuel + maintenance cost vs revenue)
- No early warning when a driver/vehicle’s fuel efficiency degrades — often the first sign of engine wear, tyre problems, or fuel pilferage

3. Product Vision & Differentiation

3.1 Vision

A single system where every stakeholder — Fleet Manager, Dispatcher, Safety Officer, Financial Analyst — works off the same live data, and the system itself enforces the business rules that today rely on someone remembering them.

3.2 How this differs from Ecount-style tools

Dimension	Typical Ecount-style tool	TransitOps
Data model	Asset-centric (vehicle records, logs)	Trip-centric — every cost/event is tied back to a trip, vehicle, and driver simultaneously
Fuel tracking	Manual log entry	Log entry + auto mileage calc (km/litre) + rolling-average deviation alerts
Maintenance	Calendar/odometer-based service reminders	Service reminders plus per-tyre position-wise wear tracking and swap history
Financials	Expense logging	Expense logging auto-rolled up into per-vehicle ROI, category-wise cost breakdown, and cost-per-km
Trip patterns	Trips treated as isolated records	Trip-chain / circular-route awareness — flags vehicles that haven’t returned to home depot within an expected cycle

Access	Often single admin view	True RBAC — 4 distinct role dashboards, each scoped to what that role needs
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3.3 USP Statement

“TransitOps doesn’t just log what happened to your fleet — it tells you what it’s costing you, and warns you before it becomes a breakdown.”

4. Target Users & Personas

Role	Responsibilities	Primary Screens
Fleet Manager	Vehicle lifecycle, maintenance, tyre/asset health, acquisition cost tracking	Fleet, Vehicle Registry, Maintenance, Analytics
Dispatcher	Creates trips, assigns vehicle + driver, monitors active deliveries	Dashboard, Trip Dispatcher, Fleet (read)
Safety Officer	Driver compliance, license validity, safety scores	Drivers, Compliance alerts, Analytics (safety)
Financial Analyst	Expense review, fuel cost, maintenance cost, profitability/ROI	Fuel & Expense, Analytics, Reports export

5. Scope

5.1 In Scope (MVP — must ship in 8 hours)

- Authentication + RBAC (4 roles)
- Dashboard with KPIs and filters
- Vehicle Registry (CRUD, unique reg. no., status lifecycle)
- Driver Management (CRUD, license expiry tracking, status lifecycle)
- Trip Management with full validation lifecycle (Draft → Dispatched → Completed → Cancelled)
- Maintenance module with automatic vehicle status transitions
- Fuel & Expense logging with automatic operational cost rollup
- Reports & Analytics (fuel efficiency, utilization, cost, ROI) + CSV export
- Load type / cargo capacity / freight type fields on trips
- Category-wise expense rollup visible on each vehicle’s profile

5.2 In Scope (Enhancement — build if time allows, after MVP is stable)

- Mileage anomaly detection (rolling average + deviation alert)
- Vehicle document panel (RC, Insurance, PUC, Permit) with expiry reminders on dashboard
- Trip-chain / depot-return tracking (flag vehicles overdue for return-to-base)
- License/document expiry email or in-app notification digest

5.3 Out of Scope / Stretch (post-hackathon roadmap)

- Auto-fill of vehicle details from government RC/PUC number lookup (stubbed with mock service in MVP; real integration is a roadmap item requiring paid API/govt access)
- 2D drag-and-drop tyre position management with per-tyre km-by-position tracking
- Dark mode
- PDF export of reports (CSV is mandatory; PDF is bonus per PS)
- Native mobile app

6. Functional Requirements

6.1 Authentication & RBAC

- Email + password login; session via JWT (access + refresh token)
- Role selected/assigned at account level, not selectable at login (mock login screen shows role selector for demo/testing convenience only)
- Only authenticated users can access any application route
- Failed login lockout after 5 attempts (per wireframe)
- Each role is scoped to specific navigation items and API permissions (see RBAC matrix in Design Doc §5)

6.2 Dashboard

- KPI tiles: Active Vehicles, Available Vehicles, Vehicles in Maintenance, Active Trips, Pending Trips, Drivers on Duty, Fleet Utilization %
- Filters: Vehicle Type, Status, Region
- Recent Trips table (trip id, vehicle, driver, status, ETA)
- Vehicle Status distribution (Available / On Trip / In Shop / Retired)
- Reminders panel (enhancement): upcoming license/document expiries, mileage anomaly flags, vehicles overdue for depot return

6.3 Vehicle Registry

- Fields: Registration Number (unique, validated), Name/Model, Type (Van/Truck/Mini/etc.), Max Load Capacity (kg), Odometer, Acquisition Cost, Status
- Status values: Available, On Trip, In Shop, Retired
- Enhancement — Vehicle Document sub-panel: RC Book, Insurance, PUC, Permit, each with number + expiry date; feeds the dashboard reminder panel
- Enhancement — vehicle profile shows lifetime category-wise expense totals (Fuel / Maintenance / Tolls / Misc) and computed ROI

6.4 Driver Management

- Fields: Name, License Number, License Category, License Expiry Date, Contact Number, Safety Score, Status
- Status values: Available, On Trip, Off Duty, Suspended
- Expired license or Suspended status automatically excludes driver from trip assignment pool
- Safety Officer role can adjust Safety Score and Suspended status; other roles view read-only

6.5 Trip Management

- Create trip: Source, Destination, Vehicle (available only), Driver (available only), Cargo Weight, Planned Distance
- **Enhancement fields:** Load Type (e.g., Palletized/Loose/Container), Freight Type (what is being carried — e.g., FMCG, Industrial, Perishable), Declared Cargo Capacity used (%)
- Trip lifecycle: Draft → Dispatched → Completed → Cancelled
- Live board shows trips grouped by lifecycle stage
- On Completion: capture final odometer + fuel consumed → auto-creates a fuel log entry tied to that trip
- **Enhancement — Trip chaining:** system tracks the sequence of trips per vehicle; if a vehicle has not returned to its home depot within a configurable number of trips/days, it is flagged on the dashboard reminder panel

6.6 Maintenance

- Create maintenance record: Vehicle, Service Type, Cost, Date, Status (In Shop / Completed)
- Creating an active maintenance record automatically sets vehicle status to In Shop and removes it from dispatch

pool

- Closing maintenance restores vehicle to Available (unless vehicle is Retired)
- **Enhancement (stretch) — Tyre Management sub-module:**
 - Register tyres per vehicle with position (e.g., Front-Left, Front-Right, Rear-Left-Outer, Rear-Left-Inner, Rear-Right-Outer, Rear-Right-Inner — configurable by axle count)
 - Each tyre tracks cumulative km run at each position it has occupied
 - Position changes (swap/rotation) logged with date + odometer reading, preserving km history per position
 - 2D visual truck layout with drag-and-drop to reassign tyre position (stretch UI goal; can ship as a simple table-based position editor if time is short)
 - Wear analytics: flags tyres that reached replacement threshold faster than fleet average

6.7 Fuel & Expense Management

- Fuel log: Vehicle, Date, Litres, Cost — can be entered standalone or auto-created from trip completion
- Other expenses: Toll, Misc, tied to a trip or vehicle
- **Enhancement — Mileage intelligence:** every fuel log captures odometer-at-refuel; system computes km/litre for that fill-up and updates the vehicle's rolling average mileage
 - If a fill-up's mileage falls more than a configurable threshold (e.g., 15%) below the vehicle's (or that driver's) rolling average across the last N fill-ups, the system raises a **Mileage Deviation Alert** on the dashboard and vehicle profile
- Total Operational Cost (Auto) = Fuel + Maintenance, computed per vehicle
- **Enhancement:** every expense (fuel, toll, maintenance, misc) rolls up into the vehicle's profile as category-wise lifetime totals, visible in Reports & the vehicle detail view

6.8 Reports & Analytics

- Fuel Efficiency (Distance / Fuel, km/l)
- Fleet Utilization %
- Operational Cost (aggregate and per-vehicle)
- Vehicle ROI = $(\text{Revenue} - (\text{Maintenance} + \text{Fuel})) / \text{Acquisition Cost}$
- Monthly Revenue trend chart
- Top Costliest Vehicles chart
- **Enhancement:** Category-wise expense breakdown per vehicle (pie/bar), Mileage trend line with anomaly markers, Tyre wear summary (if tyre module is built)
- CSV export (mandatory); PDF export (bonus)

6.9 Settings & RBAC

- General depot settings: name, currency, distance unit
- Role-based access matrix editable by admin/Fleet Manager (view-only for other roles)

7. Business Rules (Mandatory + Derived)

1. Vehicle registration number must be unique.
2. Retired or In Shop vehicles never appear in the dispatch/vehicle-selection pool.
3. Drivers with expired licenses or Suspended status cannot be assigned to trips.
4. A driver or vehicle already marked On Trip cannot be assigned to another trip.
5. Cargo Weight must not exceed the vehicle's maximum load capacity — dispatch is blocked otherwise, with a clear validation message (as shown in wireframe).
6. Dispatching a trip automatically sets both vehicle and driver status to On Trip.
7. Completing a trip automatically sets both vehicle and driver status back to Available, and creates the associated fuel log entry.
8. Cancelling a dispatched trip restores vehicle and driver to Available.
9. Creating an active maintenance record automatically sets vehicle status to In Shop.

- 10. Closing maintenance restores vehicle to Available, unless the vehicle is Retired.
- 11. *(Enhancement)* A fuel log entry more than [threshold]% below the vehicle’s rolling average mileage triggers a Mileage Deviation Alert.
- 12. *(Enhancement)* A vehicle not returned to its designated home depot within [N] trips or [N] days triggers a Depot Return Overdue flag.
- 13. *(Enhancement)* A document (RC/Insurance/PUC/Permit) within [N] days of expiry triggers a reminder on the dashboard; an expired document blocks the vehicle from dispatch (configurable — off by default for MVP demo to avoid blocking test data).
- 14. *(Stretch)* A tyre that reaches its position-specific wear threshold before the fleet-average km is flagged for early-replacement review.

8. Non-Functional Requirements

- Responsive web UI (desktop-first, usable on tablet)
- API response time target: < 500ms for standard CRUD/list endpoints
- Role-based authorization enforced at the API layer, not just hidden in UI
- Data integrity: all status transitions happen via transactional service-layer calls, never direct field edits, to prevent inconsistent states
- Auditable: key transitions (dispatch, complete, cancel, maintenance open/close) timestamped and attributable to the acting user

9. Data Entities (Overview)

Users, Roles, Vehicles, Vehicle Documents, Drivers, Trips, Trip Expenses (Fuel Logs), Other Expenses, Maintenance Records, Tyres, Tyre Position History, Notifications/Reminders.

Full schema with fields, keys, and relationships is defined in the **TransitOps Design Document, Section 3**.

10. Acceptance Criteria (Sample — Core Flow)

Given the example workflow in the problem statement:

1. Register vehicle ‘Van-05’ (capacity 500kg, status Available) → appears in Vehicle Registry, unique reg. no. enforced.
2. Register driver ‘Alex’ with valid license → appears in Available drivers pool.
3. Create trip, Cargo Weight 450kg → system validates $450 \leq 500$ → dispatch allowed.
4. Dispatch → Vehicle & Driver status become On Trip; both disappear from other trips’ selection pools.
5. Complete trip (final odometer + fuel consumed) → Vehicle & Driver return to Available; fuel log auto-created; mileage computed.
6. Create maintenance record (Oil Change) → Vehicle status becomes In Shop, hidden from dispatch.
7. Close maintenance → Vehicle returns to Available.
8. Reports reflect updated operational cost and fuel efficiency immediately.

Each step above must be independently testable and demoable.

11. MVP Prioritization (MoSCoW — for the 8-hour build)

Priority	Items
Must	Auth+RBAC, Vehicle Registry CRUD, Driver CRUD, Trip lifecycle + all 8 mandatory business rules, Maintenance status automation, Fuel/Expense logging + operational cost calc, Dashboard KPIs, Reports (4 core metrics) + CSV export

Should	Load type/freight type fields, category-wise expense rollup on vehicle profile, mileage deviation alert (basic version), filters/search across modules
Could	Vehicle document panel + expiry reminders, depot-return-overdue flag, charts polish (top costliest vehicles, monthly revenue)
Won't (this hackathon)	Tyre 2D drag-drop module, real RC/PUC government lookup, dark mode, PDF export, mobile app, email delivery (in-app notification only)

12. Risks & Mitigations

Risk	Mitigation
8-hour window too tight for full stack + all enhancements	Strict MoSCoW cut line; Must-list is independently demoable end-to-end
Real vehicle-document lookup APIs require paid access/KYC	Stub with a mock lookup service returning sample data, documented as a roadmap item
Tyre 2D UI is complex for remaining time	Ship as simple position-select dropdown + history table if drag-drop isn't feasible
Business-rule bugs (e.g., race condition on double-dispatch)	Enforce all transitions via atomic DB transactions in the service layer, not client-side checks alone